Grade 4

FAST Mathematics

Sample Test Materials

The purpose of these sample test materials is to orient teachers and students to the types of paper-based FAST Mathematics questions. By using these materials, students will become familiar with the types of items and response formats they may see on a paper-based test. The sample items and answers are not intended to demonstrate the length of the actual test, nor should student responses be used as an indicator of student performance on the actual test. The sample test materials are not intended to guide classroom instruction.

Grade 4 FAST Mathematics Reference Sheet

DO NOT REMOVE THIS SHEET

Customary Conversions

1 foot = 12 inches
1 yard = 3 feet

1 pint = 2 cups
1 quart = 2 pints
1 gallon = 4 quarts

1 pound = 16 ounces

Metric Conversions

1 meter = 100 centimeters
1 meter = 1000 millimeters
1 kilometer = 1000 meters

1 liter = 1000 milliliters

1 gram = 1000 milligrams
1 kilogram = 1000 grams

Time Conversions

1 minute = 60 seconds
1 hour = 60 minutes

Formulas

Rectangle $P = l + l + w + w$
$A = l \times w$

Key	
l = length w = width	P = perimeter A = area

Use the space in this Test and Response Book to do your work. Then, completely fill in the bubble beside the answer you choose. For some items, filling in more than one bubble may be required, so read each item carefully. If you change your answer, be sure to erase completely.

Some items will ask you to write a response in a shaded box or boxes. See the sample item below.

Sample Item:

What number is one-hundredth more than 732.12?

Write your response in the shaded box below.

Write your response in the box.

Some items may have more than one box, so read each item carefully. Your answers for the items with response boxes may contain whole numbers, fractions, or decimals.

1. What number is one-hundredth more than 732.12?

Write your response in the shaded box below.

2. A list of numbers is shown.

419,572
431,257
413,725

Select numbers to order them from greatest to least. For each box, fill in the bubble before the number that is correct.

	;		;	
Ⓐ 419,572 Ⓑ 431,257 Ⓒ 413,725	;	Ⓐ 419,572 Ⓑ 431,257 Ⓒ 413,725	;	Ⓐ 419,572 Ⓑ 431,257 Ⓒ 413,725

3. Hanson adds cups of flour to a bowl to make dough.

- He needs $8\frac{3}{5}$ cups of flour.
- He has already added $6\frac{2}{5}$ cups.

How many more cups of flour does Hanson need to add?

Ⓐ $1\frac{1}{5}$

Ⓑ $2\frac{1}{5}$

Ⓒ $2\frac{5}{10}$

Ⓓ $14\frac{5}{10}$

4. Marcy draws a rectangle.

- The rectangle has an area of 12 square inches and a perimeter of 16 inches.
- Marcy wants to draw a second rectangle with the same perimeter but a different area.

What are possible values, in inches, for the length and width of the second rectangle?

Write your responses in the shaded boxes below.

Length:

Width:

5. Hannah has 3 baseballs. Each baseball weighs $\frac{5}{16}$ pound.

Select all the expressions that represent the total weight, in pounds, of all 3 baseballs.

Ⓐ $\frac{5}{16} + 3$

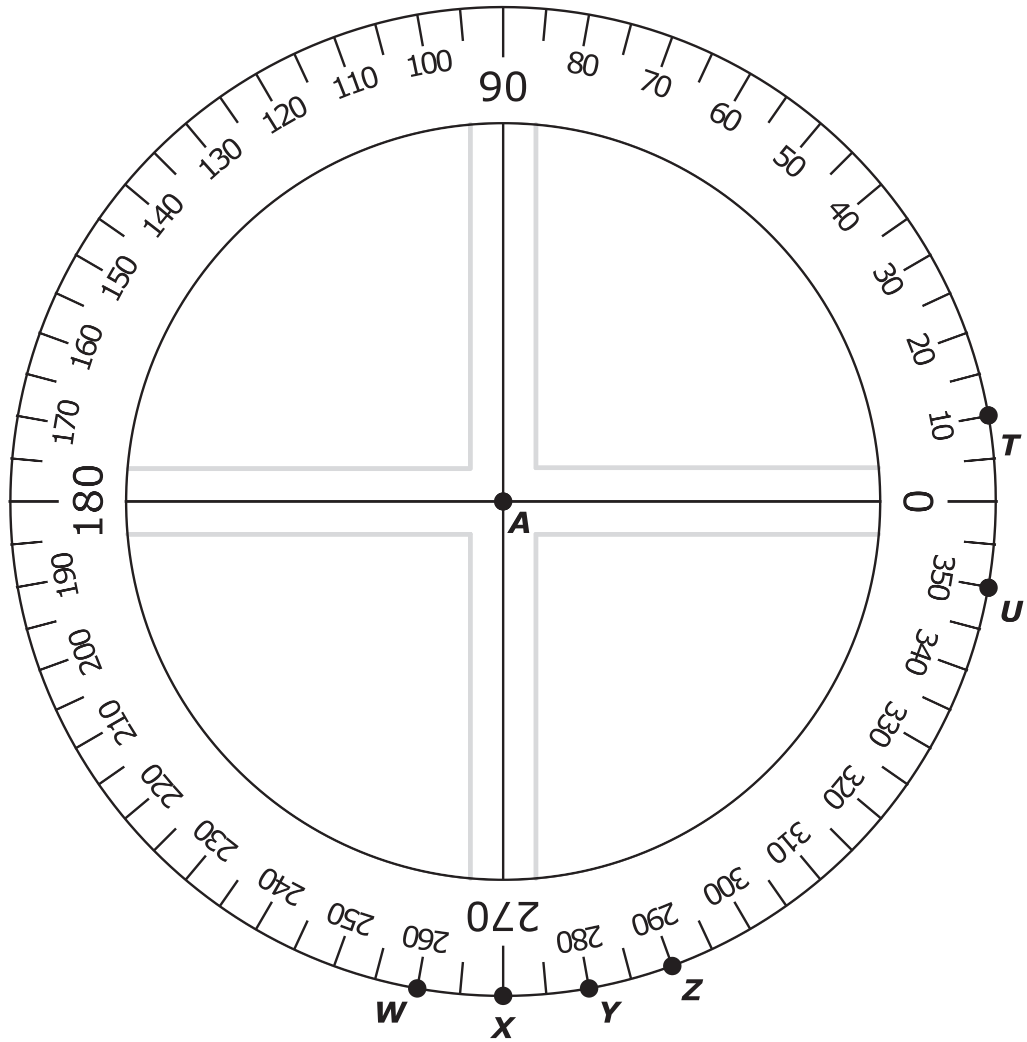
Ⓑ $\frac{5}{16} \times 3$

Ⓒ $\frac{5}{16} \times \frac{3}{1}$

Ⓓ $\frac{5}{16} \times \frac{3}{3}$

Ⓔ $\frac{5}{16} + \frac{5}{16} + \frac{5}{16}$

6. A protractor with labeled points is shown.



Select two points to draw two rays to form an angle measuring 80 degrees. For each blank, fill in the bubble **before** the point that is correct.

Draw a ray from point *A* to point __________ [Ⓐ *T* Ⓑ *U*].

Draw a second ray from point *A* to point __________ [Ⓐ *W* Ⓑ *X* Ⓒ *Y* Ⓓ *Z*].

7. Fill in bubbles to match each decimal with all its equivalent fractions.

	$\frac{9}{10}$	$\frac{90}{100}$	$\frac{9}{100}$
0.9	Ⓐ	Ⓑ	Ⓒ
0.09	Ⓓ	Ⓔ	Ⓕ

Go On

8. This question has **two** parts.

Part A

Fill in the bubble to select the correct comparison of the two decimals.

Ⓐ $237.1 < 237.04$
Ⓑ $237.1 > 237.04$

Part B

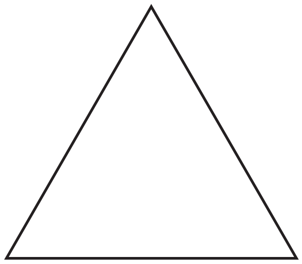
Which statement explains why the comparison is true?

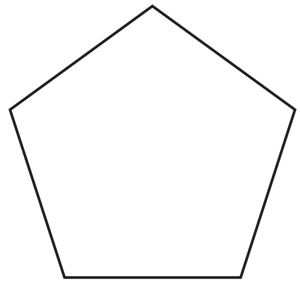
- Ⓐ One-tenth is less than four-hundredths.
- Ⓑ One-tenth is greater than four-hundredths.
- Ⓒ One-hundredth is less than four-tenths.
- Ⓓ One-hundredth is greater than four-tenths.

9. What is the value of $3\frac{3}{8} - 2\frac{6}{8}$?

Write your response in the shaded box below.

10. Select all the shapes that have at least one acute angle.

A


C


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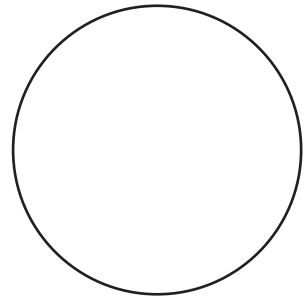
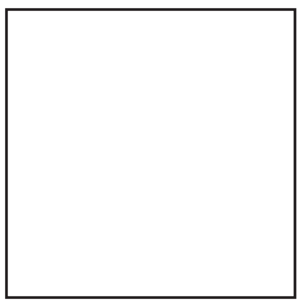
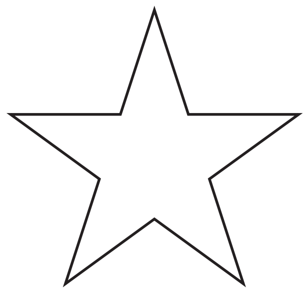


B


D


FLORIDA ASSESSMENT OF STUDENT THINKING

Grade 4

FAST Mathematics

Sample Items Answer Key

The Grade 4 FAST Mathematics Sample Items Answer Key provides the correct response(s) for each item on the sample test. The sample items and answers are not intended to demonstrate the length of the actual test, nor should student responses be used as an indicator of student performance on the actual test.

1. What number is one-hundredth more than 732.12?

Other correct responses: *any equivalent value*

Go On

2. A list of numbers is shown.

419,572
431,257
413,725

Select numbers to order them from greatest to least.

431,257 ; 419,572 ; 413,725

3. Hanson adds cups of flour to a bowl to make dough.

- He needs $8\frac{3}{5}$ cups of flour.
- He has already added $6\frac{2}{5}$ cups.

How many more cups of flour does Hanson need to add?

- Ⓐ $1\frac{1}{5}$
- ● $2\frac{1}{5}$
- Ⓒ $2\frac{5}{10}$
- Ⓓ $14\frac{5}{10}$

Option B: **This answer is correct.** *The student correctly subtracted* $6\frac{2}{5}$ *from* $8\frac{3}{5}$.

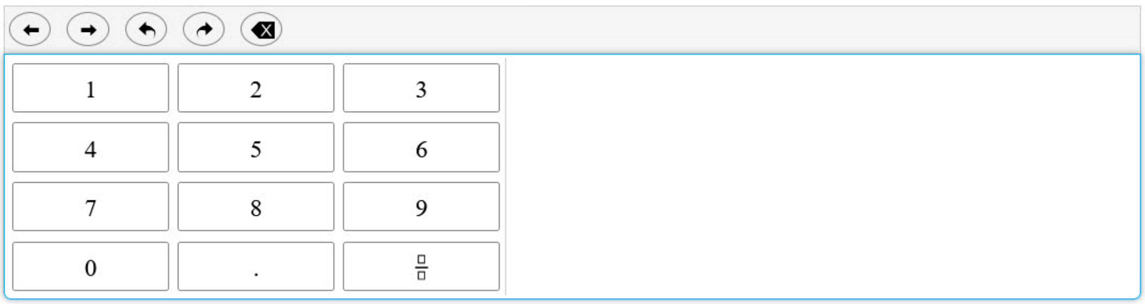
4. Marcy draws a rectangle.

- The rectangle has an area of 12 square inches and a perimeter of 16 inches.
- Marcy wants to draw a second rectangle with the same perimeter but a different area.

What are possible values, in inches, for the length and width of the second rectangle?

$Length$: 5

$Width$: 3



Other correct responses: *any equivalent value*

Note: *The values for length and width can be reversed.*

5. Hannah has 3 baseballs. Each baseball weighs $\frac{5}{16}$ pound.

Select all the expressions that represent the total weight, in pounds, of all 3 baseballs.

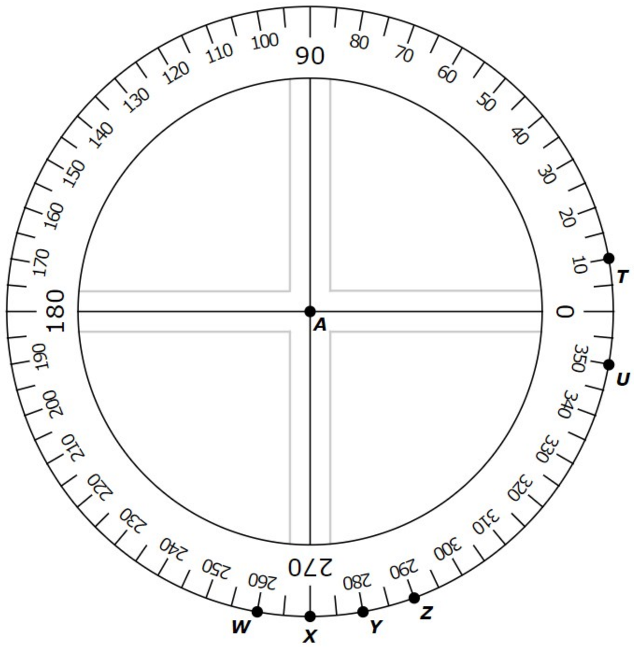
- [ ] $\frac{5}{16} + 3$
- [x] $\frac{5}{16} \times 3$
- [x] $\frac{5}{16} \times \frac{3}{1}$
- [ ] $\frac{5}{16} \times \frac{3}{3}$
- [x] $\frac{5}{16} + \frac{5}{16} + \frac{5}{16}$

<u>*Option B*</u>*:* **This answer is correct.** *The student correctly identified that 5/16 × 3 accurately represents the situation.*

<u>*Option C*</u>*:* **This answer is correct.** *The student correctly identified that 5/16 × 3/1 accurately represents the situation.*

<u>*Option E*</u>*:* **This answer is correct.** *The student correctly identified that the numerator of the expression 5/16 can be added to itself 3 times to find the total weight of 3 baseballs.*

6. A protractor with labeled points is shown.



Select two points to draw two rays to form an angle measuring 80 degrees.

Draw a ray from point *A* to point *T*.

Draw a second ray from point *A* to point *Z*.

Other correct responses:

Select two points to draw two rays to form an angle measuring 80 degrees.

Draw a ray from point *A* to point *U*.

Draw a second ray from point *A* to point *X*.

7. Match each decimal with all its equivalent fractions.

	$\frac{9}{10}$	$\frac{90}{100}$	$\frac{9}{100}$
0.9	☑	☑	☐
0.09	☐	☐	☑

8. This question has **two** parts.

Part A
Select the correct comparison of the two decimals.

237.1 > 237.04

Part B
Which statement explains why the comparison is true?

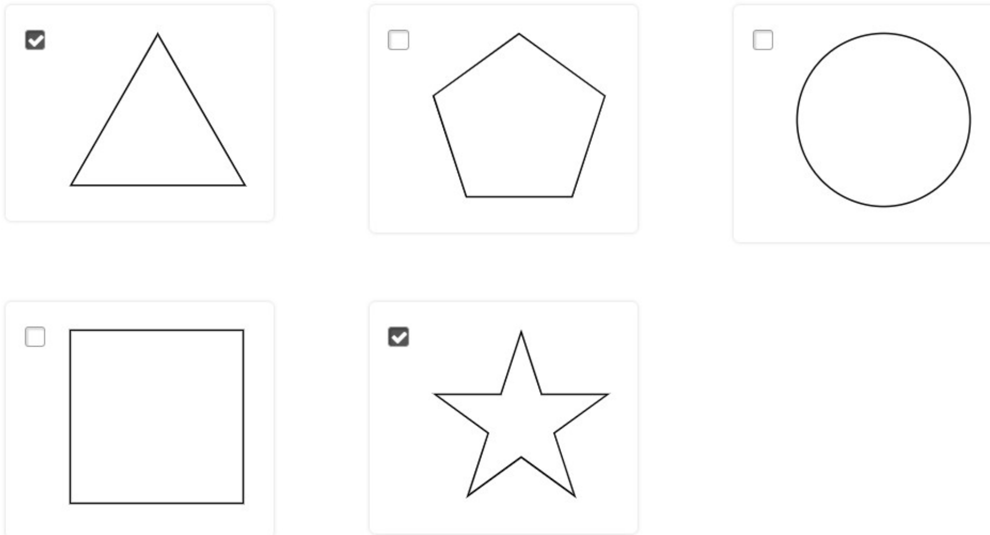
- Ⓐ One-tenth is less than four-hundredths.
- ● One-tenth is greater than four-hundredths.
- Ⓒ One-hundredth is less than four-tenths.
- Ⓓ One-hundredth is greater than four-tenths.

Option B: **This answer is correct.** *The student identified that one-tenth is larger than four-hundredths.*

9. What is the value of $3\frac{3}{8} - 2\frac{6}{8}$?

Other correct responses: *any equivalent value*

10. Select all the shapes that have at least one acute angle.



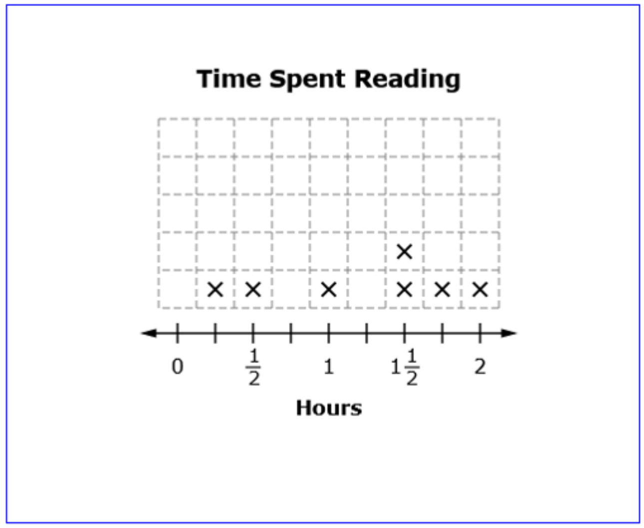
Option A: **This answer is correct.** *The student recognized that each angle in the triangle is less than 90 degrees and is therefore acute.*

Option D: **This answer is correct.** *The student recognized that the angle at each point of the star is less than 90 degrees and is therefore acute.*

11. Layla reads her book every day for a week. The amount of time she spends reading each day, in hours, is shown.

2, $\frac{1}{2}$, $1\frac{1}{2}$, $\frac{1}{4}$, 1, $1\frac{3}{4}$, $1\frac{1}{2}$

Click above the number line to create a line plot for these times.

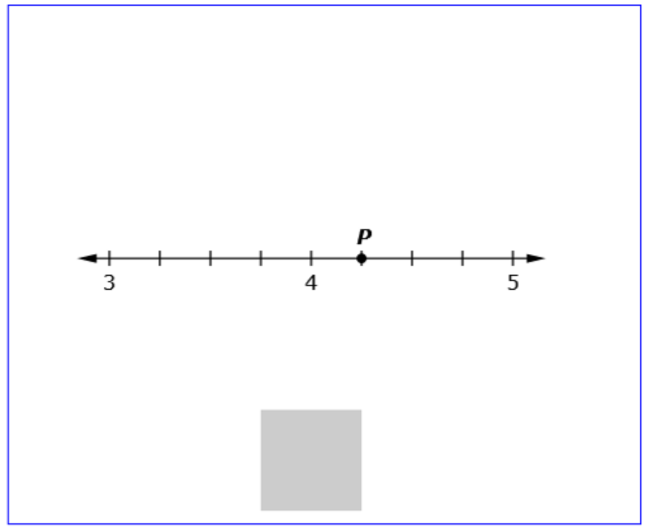


Go On

12. A list of four decimal numbers is shown.

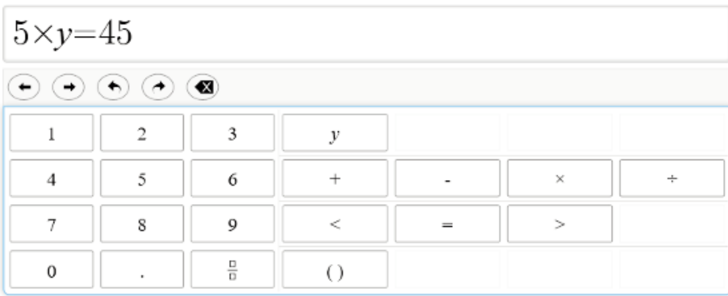
3.75, 4.25, 3.5, 4.0

Drag point *P* to the number line to represent the largest decimal in the list.



13. Vince reads *y* poems each day for 5 days. He reads a total of 45 poems.

Create an equation to represent how many poems, *y*, Vince reads each day.



Other correct responses: *any equivalent equation*

Go On

14. Amy and Gilbert ride their bikes 8 miles every hour for 3 hours. They each write an equation to determine how many miles, *m*, they ride their bikes, as shown.

Amy	Gilbert
$m \div 3 = 8$	$3 \times m = 8$

Complete the sentence to show how Amy and Gilbert can determine how many miles they ride their bikes.

They can use [only Amy's] equation to determine that they ride their bikes [24] miles.

Other correct responses: *any equivalent value of 24*

15. This question has **two** parts.

Velma wants to buy 2 books for $6.59 each and a pack of markers for $2.47.

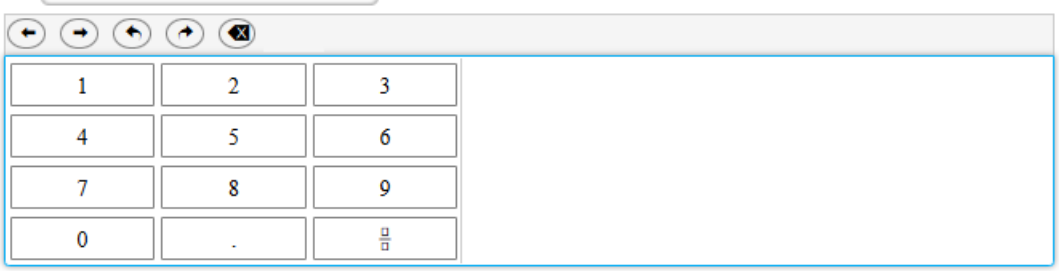
She makes an error in her work to find the total amount of money she needs to buy the items.

Part A
Click on the number in Velma's work where her error first appears.

Step	Work
Step 1	6.59 + 2.47 + 2.47
Step 2	6.59 + 4.94
Step 3	11.53

Part B
What is the correct total amount of money, in dollars, Velma needs?

$ 15.65



Other correct responses: *for Part A, either 2.47 is accepted; for Part B, any equivalent value*

16. An expression is shown.

$\frac{2}{3} + \frac{1}{3}$

Plot the value of the expression.